

COIMBRA AP, Portugal

WMO: 085480

Lat: 40.1575N

Lon: 8.4686W

Elev: 564

StdP: 14.40

Time Zone: 0.00 (EUW)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	36.4	38.7	23.1	17.8	43.4	27.3	21.7	44.8	20.0	52.2	17.6	53.4	4.1	70	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.8	92.8	68.7	88.4	67.6	84.2	66.7	71.8	85.3	70.1	82.4	68.8	80.0	7.1	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.0	105.0	74.1	66.5	99.6	73.2	65.2	95.1	72.0	35.7	85.1	34.4	82.8	33.2	80.2	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.1	13.7	11.8	32.3	99.6	2.0	3.0	30.9	101.7	29.7	103.5	28.6	105.2	27.1	107.3		
(5)			29.3	74.6	2.5	1.6	27.5	75.8	26.0	76.7	24.6	77.6	22.7	78.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.4	50.0	51.5	55.7	57.7	62.2	67.1	70.0	71.3	69.0	63.4	54.8	51.2		
	DBStd	9.09	4.45	4.66	5.45	5.74	5.90	5.62	4.90	4.88	4.81	5.34	4.89	4.57		
	HDD50	155	56	34	10	3	0	0	0	0	0	0	11	41		
	HDD65	2376	466	377	292	232	128	35	5	1	12	93	306	429		
	CDD50	3944	55	77	187	233	378	513	621	660	570	417	156	77		
	CDD65	690	0	0	4	12	41	98	162	196	131	45	1	0		
	CDH74	5326	0	0	31	107	368	835	1302	1544	892	245	2	0		
	CDH80	1961	0	0	1	16	120	324	510	613	318	59	0	0		
(14) Wind	WSAvg	5.0	5.0	5.0	5.4	5.3	5.1	5.0	5.0	4.8	4.4	5.0	4.9	5.3		
(15) Precipitation	PrecAvg	39.3	5.4	5.3	3.3	3.4	3.2	1.9	0.5	0.6	2.0	4.1	5.0	4.9		
	PrecMax	64.1	13.1	13.3	7.2	8.4	6.9	6.1	2.9	2.4	5.7	11.3	18.4	13.9		
	PrecMin	27.7	0.2	1.1	0.2	1.0	0.0	0.1	0.0	0.0	0.1	0.4	0.1	0.7		
	PrecStd	8.5	3.3	3.5	2.3	1.9	1.9	1.6	0.7	0.6	1.5	3.0	3.4	3.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.4	69.8	78.0	82.4	89.8	95.3	97.3	98.8	93.0	86.7	72.5	64.9		
		MCWB	55.0	57.0	58.1	62.4	66.8	69.1	70.3	69.8	67.5	63.1	58.5	56.9		
	2%	DB	60.6	64.9	73.3	77.0	83.0	88.5	91.3	92.4	88.2	80.0	67.8	62.0		
		MCWB	53.7	54.0	56.9	60.7	64.8	67.7	69.2	68.5	66.2	63.3	58.3	56.8		
	5%	DB	58.5	61.7	69.3	72.3	77.5	82.9	85.4	86.5	83.3	75.8	64.6	60.1		
		MCWB	53.8	53.2	55.8	59.5	63.3	67.1	68.4	67.8	65.6	62.4	58.5	55.8		
	10%	DB	56.8	58.9	65.0	67.7	72.7	78.0	80.7	81.7	78.8	72.0	62.2	58.1		
		MCWB	53.2	52.4	54.5	57.7	61.7	65.7	67.3	66.9	65.0	62.1	57.6	54.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.6	59.3	61.7	65.9	70.4	72.8	73.4	74.0	71.0	69.4	63.9	60.3		
		MCDB	59.9	64.5	70.8	76.2	82.5	89.2	88.7	90.2	82.7	73.7	65.6	61.6		
	2%	WB	56.6	56.8	59.2	62.8	66.8	70.2	71.5	71.5	69.2	67.4	61.8	58.4		
		MCDB	58.0	61.0	66.5	73.1	78.2	84.0	85.4	84.3	80.6	72.6	64.0	59.9		
	5%	WB	55.3	55.4	57.6	60.4	64.6	68.2	69.8	69.8	67.8	65.7	60.1	56.7		
		MCDB	57.1	59.0	64.2	70.0	75.3	80.0	82.5	81.4	78.0	71.3	62.7	58.6		
	10%	WB	53.8	53.9	56.2	58.4	62.6	66.2	68.2	68.2	66.4	64.0	58.3	55.1		
		MCDB	55.9	57.7	62.4	65.6	70.9	75.9	79.5	79.0	75.4	69.3	61.0	57.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.1	13.9	15.5	15.9	17.4	19.1	20.8	21.8	19.9	15.5	12.4	11.5		
		MCDBR	13.2	17.4	22.0	23.1	25.6	27.9	29.7	29.3	27.0	21.6	14.3	12.1		
		MCWBR	8.9	10.3	11.5	11.5	12.7	12.3	11.8	11.3	10.7	9.9	8.6	8.3		
		MCDBR	9.5	13.7	17.3	20.7	21.9	24.5	25.4	24.0	21.4	15.8	10.4	9.5		
	5% WB	MCWBR	7.6	9.2	10.1	11.3	11.8	11.8	11.3	10.8	9.9	8.7	7.7	7.9		
(40) Clear-Sky Solar Irradiance	taub		0.323	0.336	0.368	0.368	0.375	0.379	0.368	0.384	0.373	0.360	0.333	0.320		
	taud		2.516	2.473	2.389	2.398	2.403	2.408	2.423	2.379	2.434	2.488	2.521	2.523		
	Ebn at Noon		267	280	280	287	286	284	286	279	275	267	261	259		
	Edh at Noon		24	29	35	36	37	37	36	37	33	28	23	22		
(44) All-Sky Solar Radiation	RadAvg		642	960	1293	1673	2047	2253	2323	2098	1627	1069	716	581		
	RadStd		102	132	175	195	191	144	99	87	126	116	116	83		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	-4.61	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+103

Nomenclature: See separate page